Performance-Focused Design Exercise Packet

Part of ASME IDETC 2025 Workshop:

Adventures in Tradespace Exploration

17 August 2025

Group ID: 5050

IDEAS³ At CLEMSON

CEDAR Clemson Engineering Design Applications and Research

Version A

Round 1:

For this design exercise, you will NOT be acting as a member of your assigned working group. You will act as a team that shares the same objectives.

INSTRUCTIONS:

Working as a cross-disciplinary team, select a single solution using the performance space plots provided in the supplemental packet. You may annotate and mark the supplemental packet as you wish. Select a single vehicle from the 20 designs presented and write the selected design's ID in the space provided in the supplemental packet.

Your team wishes to maximize the following objectives:

- **A. Top Speed**
- **B. Acceleration**
- **C. Maintainability**
- **D. Rollover Stability**

*3 - 6

ID: vehicle 3

*10 - 1

* 20 - 3

* 11 - 3

* 13 - 3

*

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Annotate, mark-up, and write on this page as needed.

Post-Activity Questions:

Describe the solution the group has selected – write in the value of the design variables.

119 in	80.2 in	70.2 in	6.8 in	543 lb-ft	4.67:1	48 in	1560 kWh	0.280 in	0.18 in

Do you think you could have found a better solution given more time?

* No because based on only the information we are given, we are limited on what we know regardless of time. Vehicle 3 occured the bottom left corner of all 6 graphs, making it the most well-rounded to achieve the desired parameters

What, in this round, stopped you from finding a better solution?

limited Information, such as cost, pictures, etc.

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Round 2:

For this design exercise, you will be acting as a member of your assigned working group, and you will be provided cross-disciplinary performance space plots.

INSTRUCTIONS:

Working in your assigned working group, select a single solution using the performance space plots provided in the supplemental packet. You may annotate and mark the supplemental packet as you wish. You wish to maximize the vehicle's performance in your own objectives, and the other working group wishes to maximize their own objectives, but you must select a single vehicle from the 20 designs given in the supplemental packet. Write the selected design's ID in the space provided in the supplemental packet.

The Suspension and Driveline working group wishes to maximize the following objectives:

E. Off Road Ability
F. Towing Capacity

\#2

The Chassis and Cabin working group wishes to maximize the following objectives:

G. Low Speed Maneuverability
H. Occupant Protection

\#19

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Post-Activity Questions:

Describe the solution the group has selected – write in the value of the design variables.

110in	80.1in	65.5in	20.8in	421 lbft	2:15:1	49 in	1660 kwh	0.939in	1.48

Do you think you could have found a better solution given more time?

Maybe since 1a & 2 were both strong options. We only chose 2 since it has greater protection, given its for the Army.

What, in this round, stopped you from finding a better solution?

limited information, like round #1,

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Round 3:

For this design exercise, you will be acting as a member of your assigned working group – you will have differing objectives and be provided performance space plots that DO NOT have any cross-disciplinary objective pairings.

INSTRUCTIONS:

Working in your assigned working group, select a single solution using the performance space plots provided in the supplemental packet. You may annotate and mark the supplemental packet as you wish. You wish to maximize the vehicle's performance in your own objectives, and the other working group wishes to maximize their own objectives, but you must select a single vehicle from the 20 designs given in the supplemental packet. Write the selected design's ID in the space provided in the supplemental packet.

The Suspension and Driveline working group wishes to maximize the following objectives:

I. **Acceleration**
J. **Operational Range**

The Chassis and Cabin working group wishes to maximize the following objectives:

K. **Maintainability**
L. **Passenger Cabin Space**

13

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Post-Activity Questions:

Describe the solution the group has selected – write in the value of the design variables.

107 in	98.2 in	66.6 in	23.1 in	512 lb-ft	5.32:1	30 in	15.36 kWh	0.284 in	0.84 in

Do you think you could have found a better solution given more time?

No because 13 was the only one close the the origin for both plots. It clearly stands out.

What, in this round, stopped you from finding a better solution?

limited information

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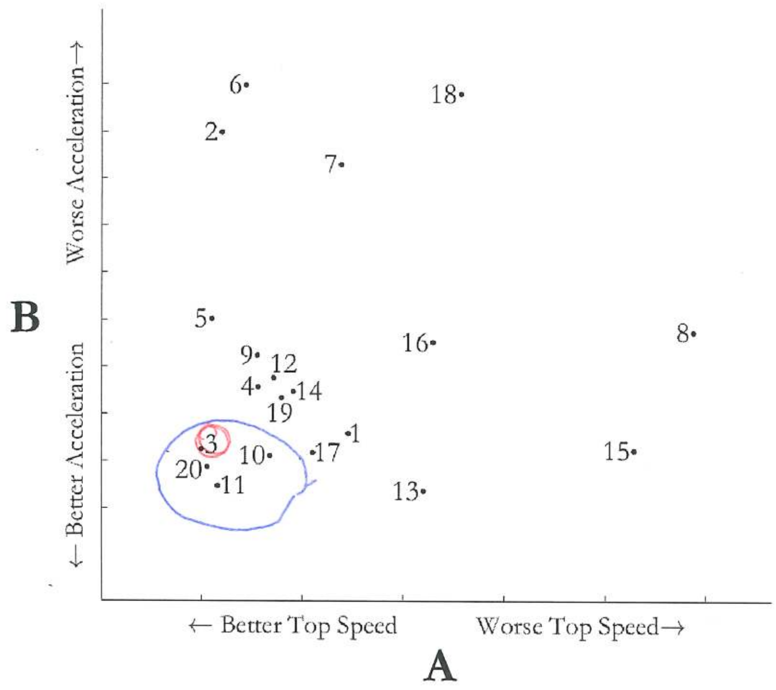

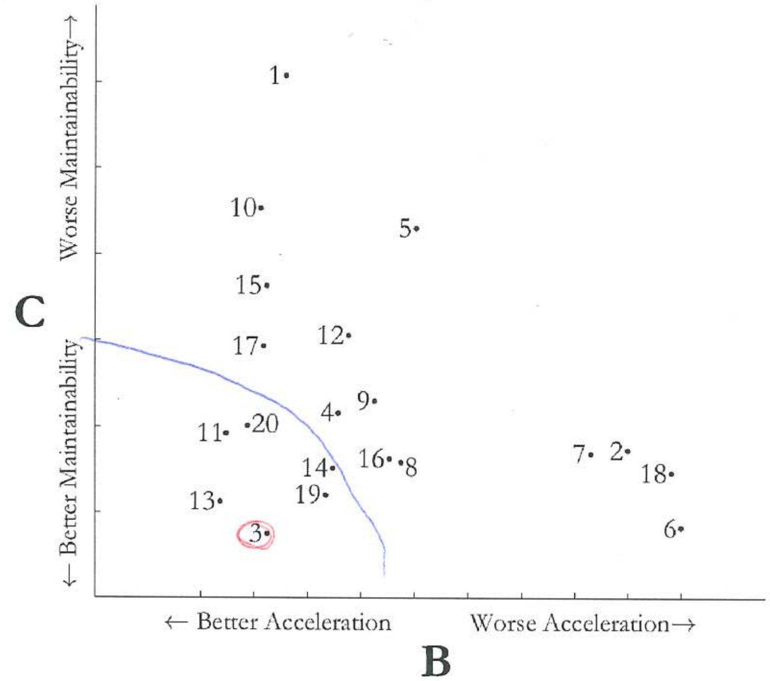

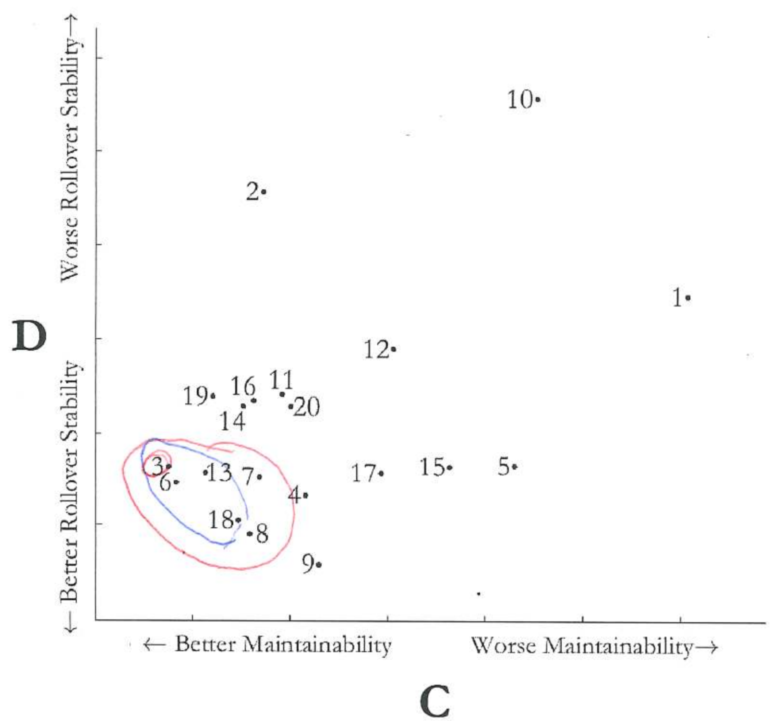

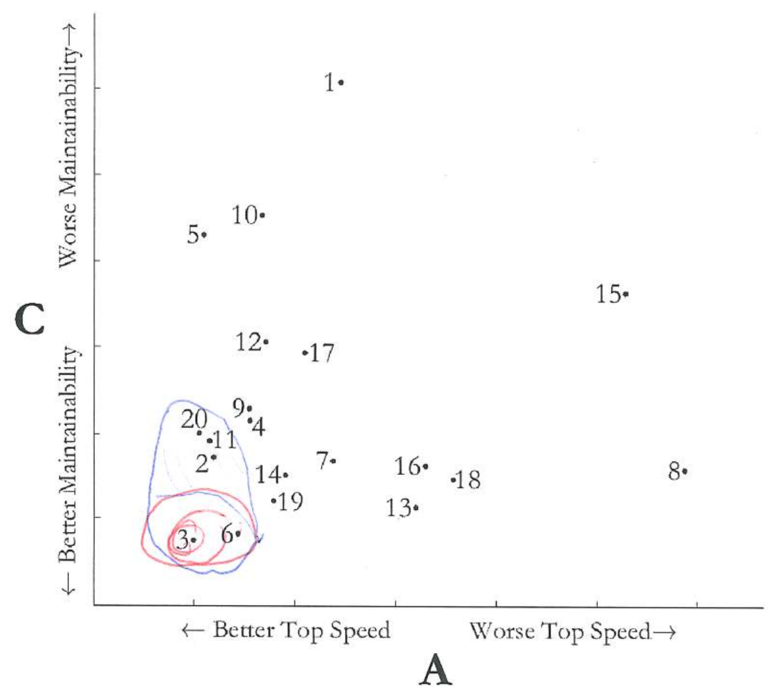

Version A

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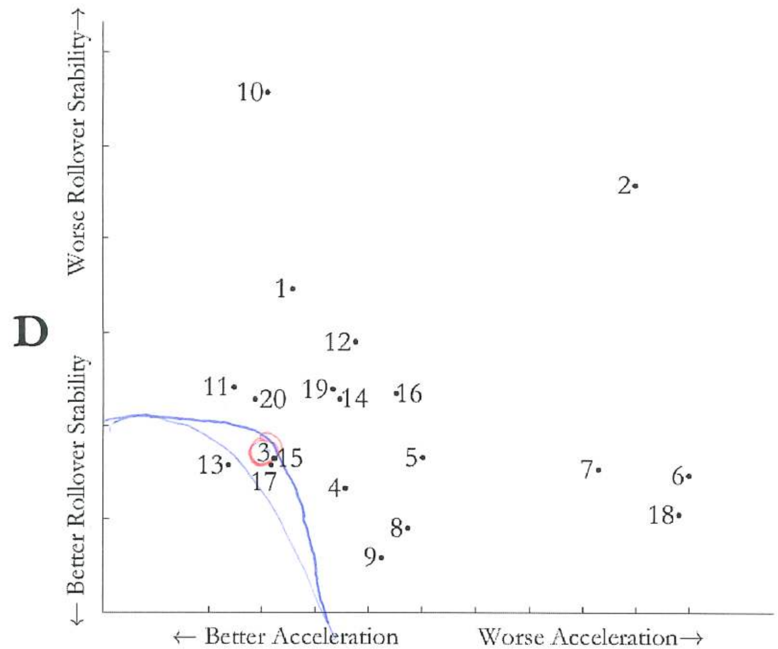

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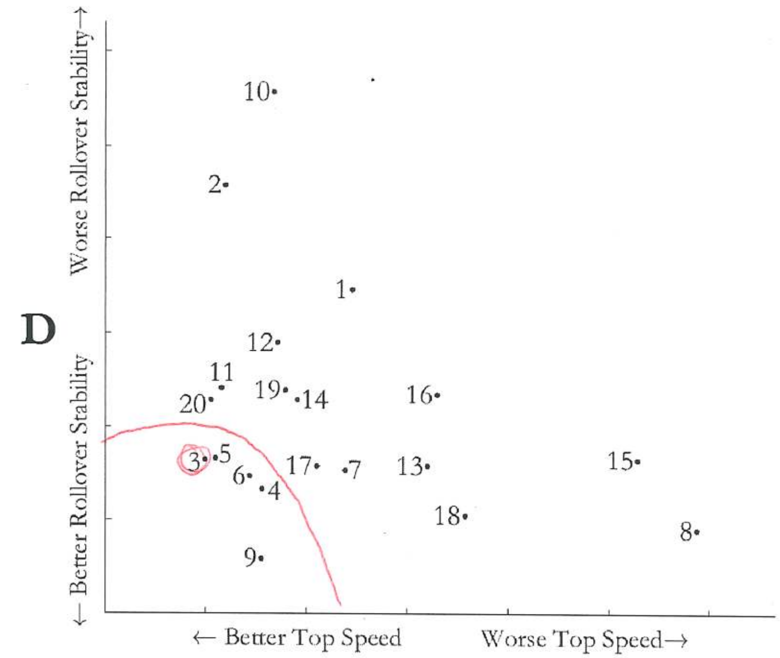

A

Enter the number ID of the selected design: 3

Write your Group ID: 5050

Version A

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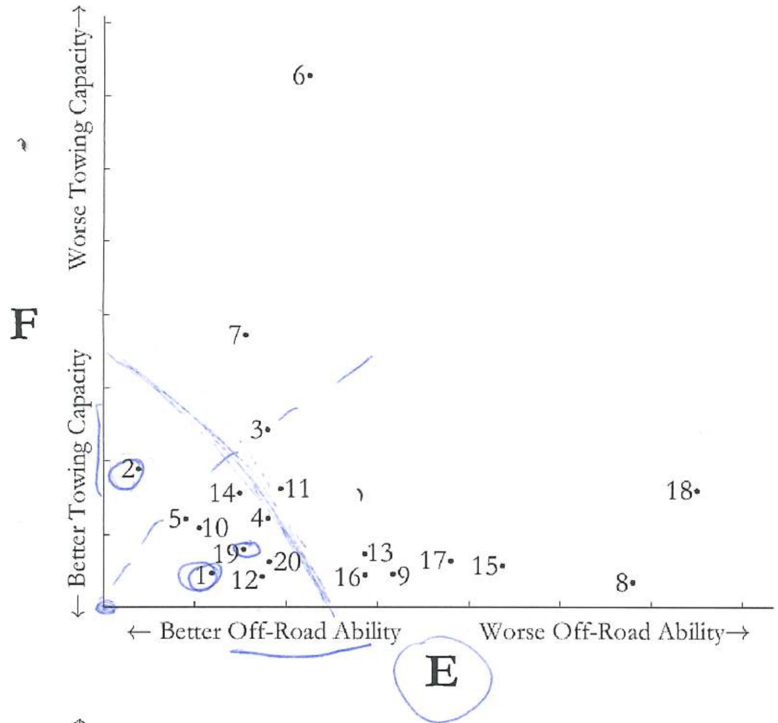

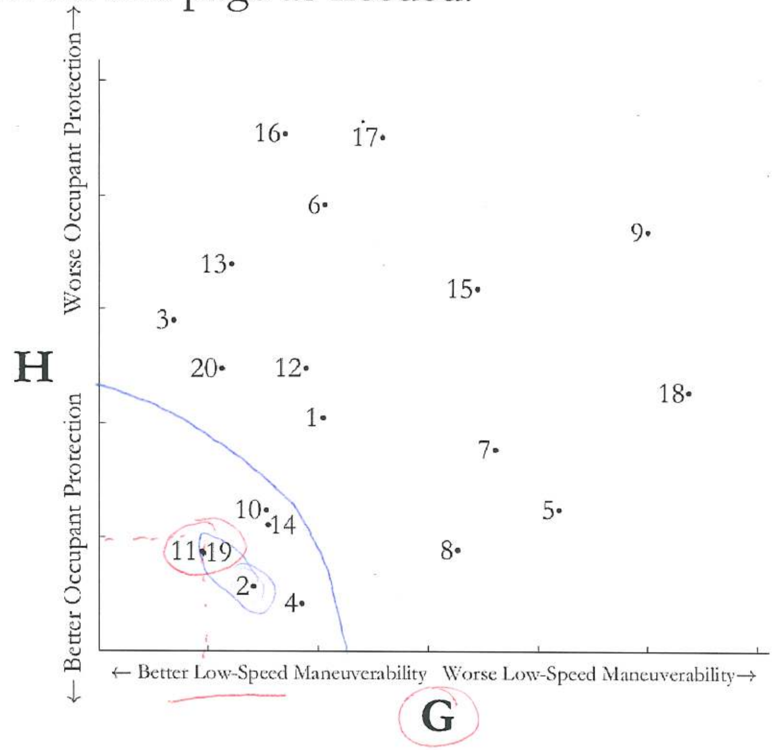

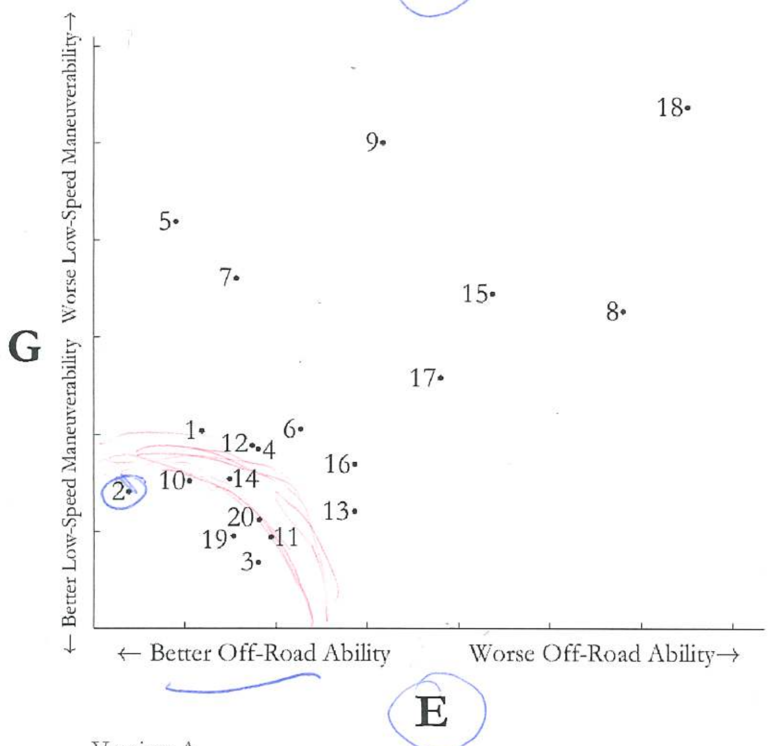

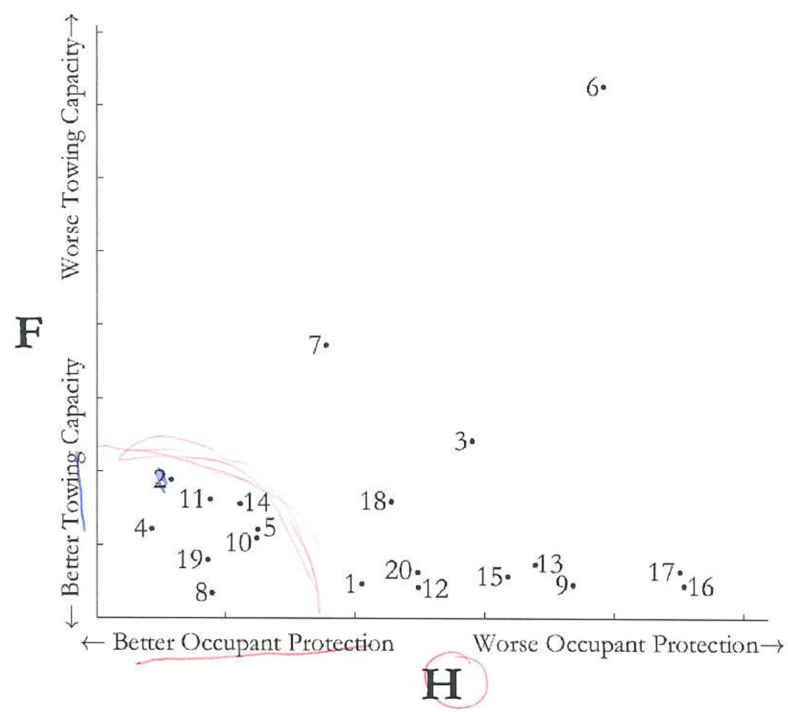

Version A

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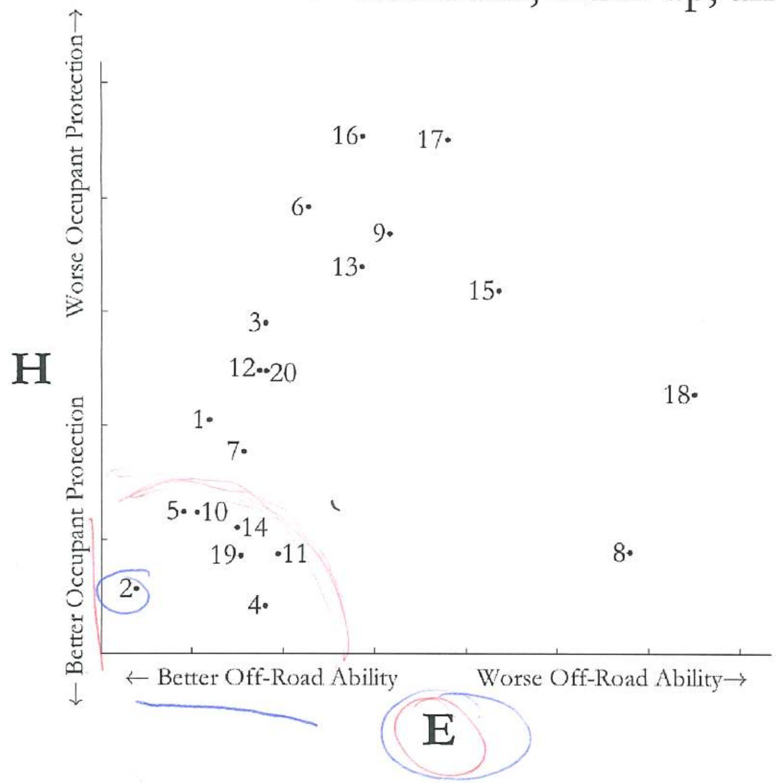

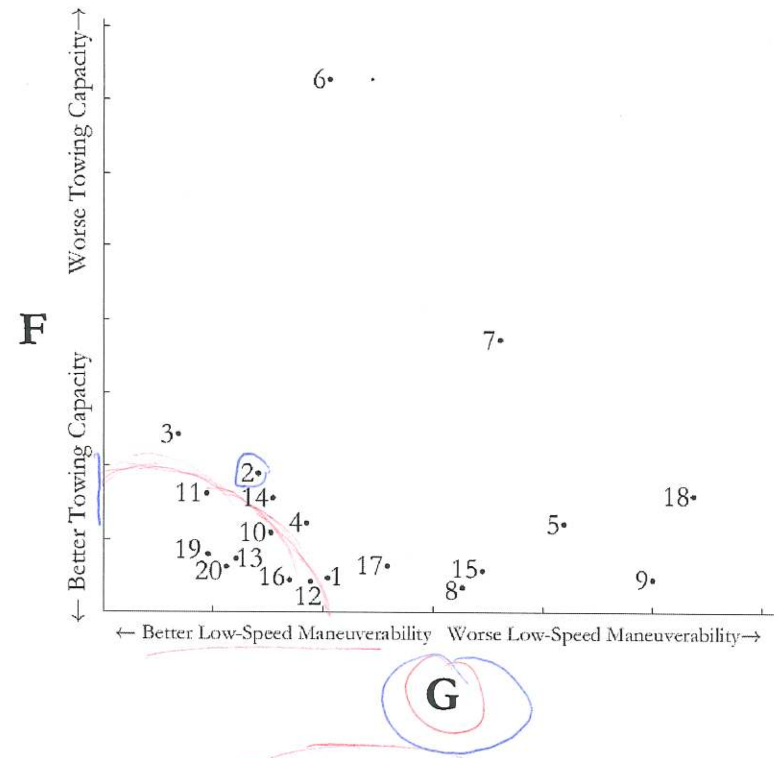

* 19 – 4/4

* 11 – 4/4

* 10 – 4/4

Enter the number ID of the selected design:______

Write your Group ID:______

Version A

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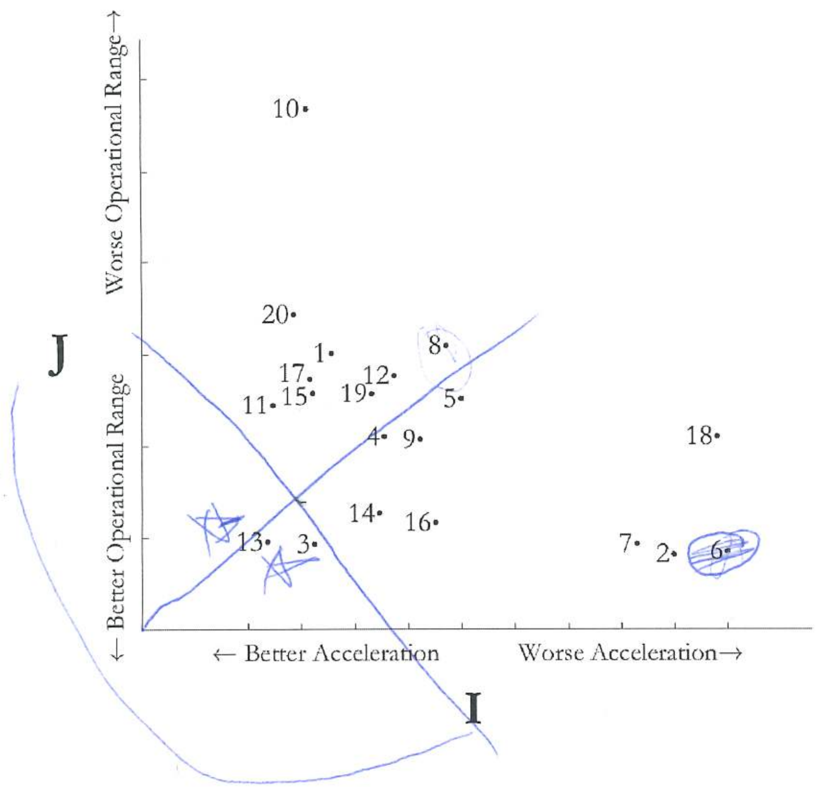

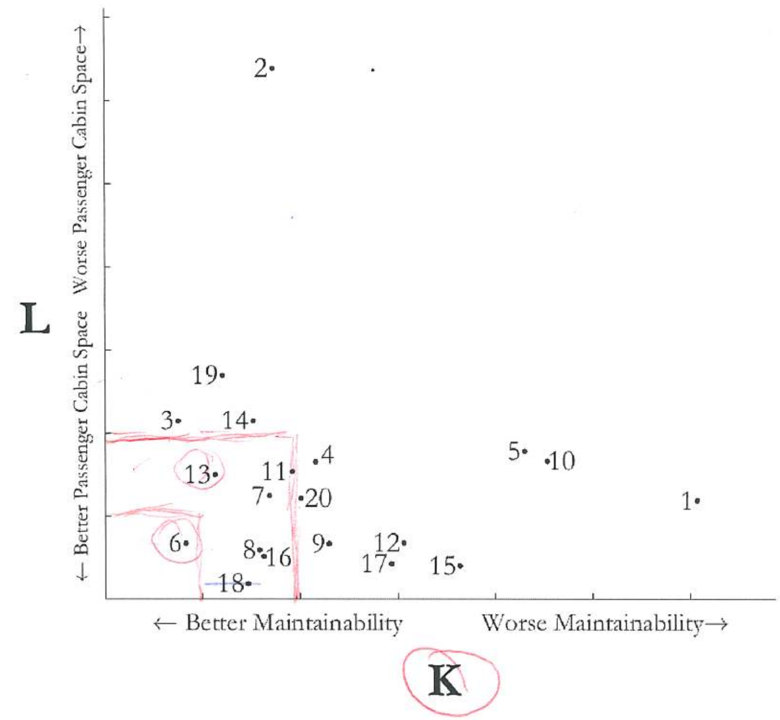

Enter the number ID of the selected design: 13

Write your Group ID: 5050

Version A